

STAT 301 In-class Activity No. 3

Consider the “wage” dataset in Activity No. 1 again. We fitted a simple linear regression model, with WAGE as response and EDUCATION as a predictor, in Activities No. 1 and 2. Is WAGE associated with both EDUCATION and SEX? Answer questions (a) and (b) below based on a multiple linear regression model.

(a) Compared with the results in activities No. 2, has the strength of the relationship between WAGE and EDUCATION changed after adjusted for SEX (i.e., after incorporating SEX information)? In other words, have the 95% confidence interval for β_1 (the parameter with EDUCATION) and the corresponding p-value changed after incorporating SEX? Why?

(b) Compared with the results in activities No. 2, What is the interpretation of the parameter (slope β_1) associated with EDUCATION in this case? Is the interpretation the same as that in activities No. 2? Why?

(c) Next, suppose that we also wish to study the relationship between WAGE and OCCUPATION, ignoring other variables in the dataset for the moment. To answer this question, fit a linear regression model. Are there significant differences in WAGE among different OCCUPATION?